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22 June 1971
SUPERSEDING
MIL-D-0050074C(MU)
9 May 1968
MIL-D-50074B
1 July 1963

MILITARY SPECIFICATION

DYE, BENZANTHRONE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers benzanthrone dye, chemically known as 1,9-benz-10-anthrone, for combat use.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Film (Polyolefin).
RR-S-366 - Sieve, Test.

MILITARY

MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

FSC 6820

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Benzanthrone dye shall conform to the chemical and physical characteristics of table I when tested as specified therein.

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Table I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Purity, dry basis, percent by weight, minimum	77	4.2.4.1
Volatile matter, percent by weight, maximum	1.0	4.2.4.2
Particle size, percent by weight passing:		4.2.4.3
840 microns (No. 20), minimum	100	
297 microns (No. 50), minimum	97	
44 microns (No. 325), maximum	40	
Percent "Marcol 52" (as anti-dusting agent)	2.0 ± 0.5	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the benzanthrone dye offered for acceptance at one time which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Three representative specimens of approximately 1/4 pound each shall be randomly removed from the lot or batch of dye

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offered for acceptance and placed in separate clean, dry containers labeled to identify the lot represented.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Polyethylene liner or closure thereof not as specified (level B only)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Desiccant not as specified or missing (level B only)
(g) Marking incorrect, missing, or illegible
(h) Palletization not as specified

4.2.3.2 For test. Each specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Purity. Prepare and measure the absorbance of a standard benzanthrone dye solution (see 6.4) and specimen solution as follows: Weigh to the nearest milligram (mg) approximately 0.350 gram (g) of the dye, which has been previously dried to constant weight at 70° to 75° C, and transfer quantitatively to a 500-milliliter (ml) volumetric flask using small quantities of chloroform. Add approximately 250 ml of chloroform and thoroughly mix to dissolve the benzanthrone. Dilute with a chloroform to 500 ml and again thoroughly mix. Transfer a 5-ml aliquot to a 200-ml volumetric flask, dilute to 200 ml with chloroform, and thoroughly mix. Using a suitable spectrophotometer, measure the absorbance of the more dilute solution at 388 $\pm$ 2 millimicrons. Use the chloroform used for preparing the solutions as the reference liquid in the spectrophotometer. Cell spacers or variable path length cells may be used in lieu of the prescribed dilution procedures. Calculate the percent purity as follows:

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$$\text{Percent purity} = \frac{A\ B\ C}{DE}$$

where: A = Percent purity of standard,
B = Absorbance value of specimen,
C = Weight of standard in grams,
D = Absorbance value of standard, and
E = Weight of specimen in grams.

4.2.4.2 Volatile matter. Weigh to the nearest milligram approximately 5 g of the specimen in a tared, glass-stoppered weighing bottle. Remove the stopper and dry to constant weight at 70° to 75° C. Cool to room temperature in a desiccator and weigh. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A - B)}{W}$$

where: A = Weight of specimen and stoppered bottle before heating in grams,
B = Weight of specimen and stoppered bottle after heating in grams, and
W = Weight of specimen in grams.

4.2.4.3 Particle size. Use sieves conforming to RR-S-366. Nest the sieves in order of decreasing mesh size (with the sieve of largest mesh on top) on a receiving pan. Weigh to the nearest 0.01 g approximately 10 g of the specimen and then use one of the following procedures:

(a) Dry method. Place the weighed specimen on the top sieve and brush the material with a camel's-hair brush until no more specimen passes through the sieve. Remove the sieve and weigh the material retained. Repeat this procedure with each of the other sieves. Calculate the percent material passing through each sieve.

(b) Wet method. Mix the weighed specimen with sufficient water in a beaker by stirring with a glass rod to produce a smooth paste. Carefully wash the thoroughly wetted specimen through the sieves. (The use of a wetting agent is permitted.) Dry the sieves in an oven at 70° to 75° C and weigh the material remaining on each sieve. Calculate the percent material passing through each sieve.

4.2.4.4 Anti-dusting agent ("Marcol 52").

(a) Reference standard. Prepare a series of six standard "Marcol 52" solutions in carbon tetrachloride containing 0.20 to 0.60 g of "Marcol 52" per 100 ml of carbon tetrachloride. (This is equivalent to 1.0 to 3.0 percent "Marcol 52" per 20 g of specimen.) Determine the

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absorbance of the standard solutions at 2.310 or 3.410 ± 0.002 micrometers using an infrared spectrophotometer which has been balanced at zero or at a predetermined setting using a carbon tetrachloride blank. Prepare a calibration chart of absorbance versus percent "Marcol 52" in a 20-g specimen.

(b) Preparation of specimen. Weigh 20.000 g of the specimen into a 150-ml beaker. Add 50 ml of carbon tetrachloride and agitate for at least 1/2 hour (the specimen will not dissolve). Filter with suction. Wash the funnel and precipitate with 25 to 30 ml of carbon tetrachloride. Quantitatively transfer the filtrate to a 100-ml volumetric flask and dilute to 100 ml with carbon tetrachloride.

(c) Procedure. Determine the absorbance of the specimen solution prepared as specified in (b) at 2.310 or 3.410 ± 0.002 micrometers using the same instrument and balancing as was used in the standard solution tests. Calculate the percent "Marcol 52" from the calibration chart prepared as specified in (a). If the spectrophotometer cell thickness is such that the instrument reads off scale, further dilutions are permitted.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level B or C as specified (see 6.2).

5.1.1 Level B. Uniform quantities of no less than 150 and no more than 300 pounds ± 1/4 percent of the marked net weight of dye shall be packed in a fiber drum conforming to rule 51, section 2 of the Uniform Freight Classification for a weight limit of over 225 but not over 300 pounds, with an aluminum barrier of 0.5 mil minimum thickness incorporated into one of the inner plies of the sidewall. Each drum shall be provided with a close fitting bag liner formed from polyethylene of 4 mils minimum thickness conforming to type I, grade and finish optional of L-P-378. Seams shall be completely heat sealed and shall meet the heat-seal strength requirements of L-P-378. The filled bag shall be closed by heat sealing, tying, or knotting. A minimum of eight units of desiccant conforming to MIL-D-3464 shall be placed on top of the closed liner. Drum closure shall be fully gasketed and shall form a tight seal between the rim and the top of the drum.

5.1.2 Level C. Uniform quantities of no less than 150 and no more than 300 pounds ± 1/4 percent of the marked net weight of dye shall be packed in containers which comply with the Uniform Freight Classification Rules, the National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation. Containers shall be capable of being stacked and of supporting superimposed loads during shipment and shall assure safe delivery to destination without damage to contents. Containers shall be acceptable for shipment at the most favorable rate of the applicable regulation provided that all requirements specified herein have been met.

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5.2 Marking. Marking shall be in accordance with MIL-STD-129. In addition, each container shall be durably and legibly marked with contrasting letters and background to show the net weight of contents and the following information:

CAUTION

Avoid breathing dust or fumes.
Use with adequate ventilation.
Causes severe irritation to some people.
Avoid contact with skin or clothing.
In case of contact, flush with water.
Wash contaminated clothing before re-use.
Individuals sensitive to this product should
not continue to work with it.

5.3 Palletization. Shipping containers shall be palletized in accordancce with the applicable requirements of MIL-STD-147 except that the pallet shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. Benzanthrone dye is intended for use in the manufacture of colored signaling smoke mixtures. This dye is intended for combat use only.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Type of pallet required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Standard dye. Standard dye for the purity determination may be obtained from Commanding Officer, Edgewood Arsenal, Quality Assurance Directorate, Edgewood Arsenal, Maryland 21010.

6.5 Anti-dusting agent. "Marcol 52" is a highly refined petroleum oil obtainable from the Humble Oil and Refining Company.

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6.6 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded of "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Navy - AS

Review activities:

Army - MD, MU(FA)
Navy - AS
DSA - GS

User activity:

Army - SM

Preparing activity:

Army - MU(EA)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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